

Database Backup and Recovery Report

Task : Backup and Recovery

1. Introduction

Database backup and recovery are essential processes to ensure data safety and availability. Backup creates a copy of the database, while recovery restores it in case of failure.

2. Objective

- To demonstrate database backup
 - To restore database in case of failure
 - To ensure no data loss during the process
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3. Tools Used

- MySQL
 - SQL
 - Command Line Interface
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4. Backup Process

The backup of the database was taken using the following command:

```
mysqldump -u root -p student_db > backup.sql
```

This command exports the entire database into a SQL file named backup.sql.

5. Restore Process

The database was restored using the following command:

```
mysql -u root -p student_db < backup.sql
```

This command recreates the database from the backup file.

6. Verification

To verify successful recovery:

- Data before and after restore was compared
- Record counts were checked

- Tables were verified using SELECT queries
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7. Output

The following outputs were verified:

- Original data before backup
- Restored data after recovery

(Screenshots are included in README file)

8. Challenges Faced

- Understanding backup and restore commands
 - Simulating database failure
 - Ensuring proper command execution
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9. Conclusion

The backup and recovery process was successfully demonstrated.

The database was backed up and restored without any data loss, ensuring data integrity.
